

 POSEIDON AI-DRIVEN STUDY

PREPARED BY
Three60 Energy
PDO — PETROLEUM DEVELOPMENT OMAN

Proved Area

OWC 1642m

— STATIC MODEL BUILDING & UPDATE
Completed UG-1 w/ UG-2 or 3 Producer

Wafra Field Static Model Building & Update

POSEIDON AI-Driven Study — PDO Wafra Field, Oman

Project Overview & Field Introduction



FIELD DISCOVERY & LOCATION

Discovered in 1983, located ~20 km from Bahja in southern Oman. Fault-dependent 3-way-dip closure structure.

📅 Est. 1983

📍 ~20 km from Bahja

🏔️ 3-Way Dip Closure



RESERVOIR PROPERTIES

Upper & Middle Gharif reservoirs (Late Permian). High-quality light oil with strong porosity.

POROSITY	API GRAVITY	VISCOSITY	INIT. PRESSURE
0.25 – 0.30	40 – 55°	3 – 4.2 cP	20,000 kPa

🏠 Perm: low–multi-Darcy

🌐 Curr. P: 4,000–15,000 kPa



PRODUCTION STATUS

STOIIP 2P: [CONFIDENTIAL] MMm³ (UG: [CONF.] MMm³ | MG: [CONF.] MMm³). Field is in mature production phase with high water cut.

🛢️ Net Oil: [CONF.] m³/day

💧 BSW: 94%

☰ Total Liq: [CONF.] m³/day

🕒 Cum. Np: [CONF.] MMm³

📊 RF: [CONF.]%



WELL INVENTORY

112 wells drilled to date across producer, injector, and abandoned/LTCI categories supporting field development.

112

Total Wells Drilled

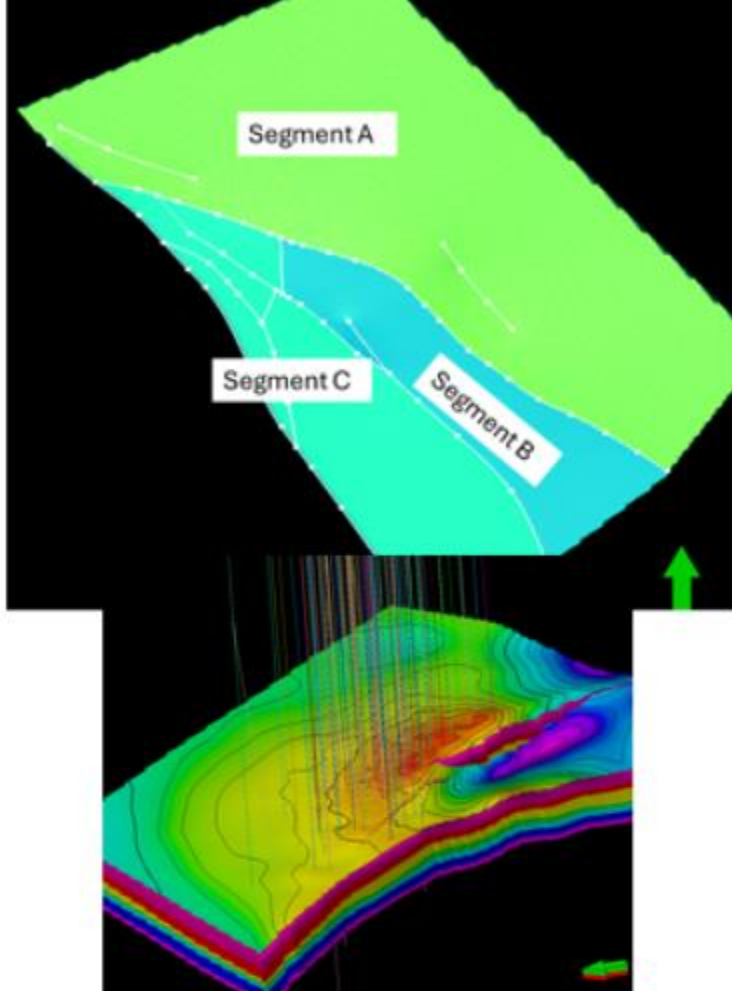
74

Operational Producers

27

Injector Wells

🚫 19 Abandoned / LTCI



3D Grid & Dimension

- Corner Point Gridding of 100mx100mx1m
- 11 fault2

8 zones
 3 segments
 175 layers
WAFRA FIELD — PETREL MODEL

3D Structural Framework Workflow — Legacy Build

• Total number of grid cells:~4.3 million

— STATIC MODEL ASSESSMENT

Legacy Petrel Model & Identified Issues



MODEL SCOPE — 2013 LEGACY BUILD

Wells **67 total** (incl. 25 new: WFA-43→67)

Base STOIIIP **[CONFIDENTIAL] MMm³**

Low **[CONF.] MMm³**

High **[CONF.] MMm³**

↓ Reduced from [CONF.] MMm³ (2010/11 study)



IDENTIFIED INCONSISTENCIES

✖ Formation Tops Mismatches

✖ Well Correlation Errors

✖ OWC Discrepancies

✖ Well Log Inconsistencies

⊘ No Uncertainty Cases Defined



AGREED REMEDIATION SCOPE

3
MONTHS

\$0
EXTRA COST

Three60 Energy agreed to execute a fast-paced static model repair at **no additional cost**, addressing all major inconsistencies in the legacy Petrel model to support the POSEIDON ROCM workflow.

Structural Framework Update

WAFRA FIELD • PDO



Fault Network Maintenance

Existing fault network retained from the legacy model to ensure structural continuity and consistency across simulation runs.



Stratigraphic Tops Update

All stratigraphic tops comprehensively reviewed and updated using new well data, improving depth accuracy across the Gharif Formation.



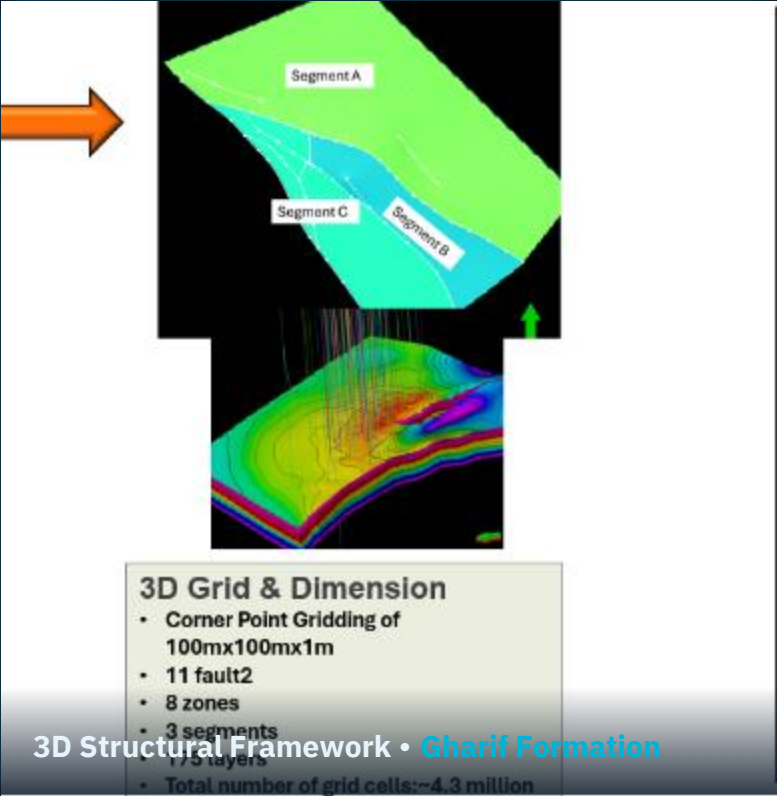
Zonation Revision

Zonation revised to sequence-package correlation (vs. legacy sand-to-sand), defining 5 Upper Gharif zones and 2 Middle Gharif zones.



Model Grid Dimensions

Model covers 10–12 km² with a 122×201×175 cell grid (~4.3 million cells). Existing layering scheme maintained for simulation consistency.



MODEL SPECIFICATIONS

~ **4.3M**

Grid Cells
(122×201×175)

10–12 km²

Model Area
Coverage

Stratigraphy & Zonation: New vs. Legacy

GHARIF FORMATION — UPPER & MIDDLE MEMBERS · WAFRA FIELD



Legacy Zonation

Sand-to-Sand Correlation

UPPER GHARIF (5 ZONES)

UG-1a	Uppermost sand unit, top of Upper Gharif	Upper
UG-1b	Secondary upper sand package	Upper
UG-2	Mid-upper sand interval	Upper
UG-3	Lower-upper sand unit	Upper
UG-4	Base of Upper Gharif sands	Upper

MIDDLE GHARIF (2 ZONES)

MG-1a	Upper Middle Gharif sand	Middle
MG-1b	Lower Middle Gharif sand	Middle

⚠ Shales excluded from sequences



New Zonation

Sequence-Package Correlation

UPPER GHARIF (5 ZONES)

HSGHU	Equiv. UG-1a · Top sequence package	Upper
HSGHUS	Equiv. UG-1b · Secondary sequence	Upper
HSGHU-2	Equiv. UG-2 · Mid-upper sequence	Upper
HSGHU-3	Equiv. UG-3 · Lower-upper sequence	Upper
HSGHU-4	Equiv. UG-4 · Base upper sequence	Upper

MIDDLE GHARIF (2 ZONES)

HSGHM-1	Equiv. MG-1a · Upper Middle sequence	Middle
HSGHM-3	Equiv. MG-1b · Lower Middle sequence	Middle

✔ Shales included in sequence packages



UPDATED



Key Improvement: New sequence-package correlation ensures geological consistency and accurate representation of vertical & lateral reservoir unit distribution — shale layers now correctly included within each sequence boundary.

Facies & Property Modelling



Facies Modelling SIS

Sequential Indicator Simulation conditioned to the RPD method (Neutron Porosity + Density). Core data from WAFRA-6; variograms based on depositional environment and FMI data.

- Clean Sandstone
- Shaly Sandstone
- Shale
- Paleosol
- Carbonate



Porosity Modelling SGS

Sequential Gaussian Simulation conditioned to facies model with normal score transform applied. Ensures porosity distribution honours lithofacies boundaries and WAFRA-6 core measurements.

- Facies-conditioned
- Normal Score Transform
- WAFRA-6 Core



Permeability Modelling SGS CO-KRIGING

SGS co-kriged with porosity to preserve the porosity–permeability relationship. Variograms derived from depositional environment interpretation and FMI image log data.

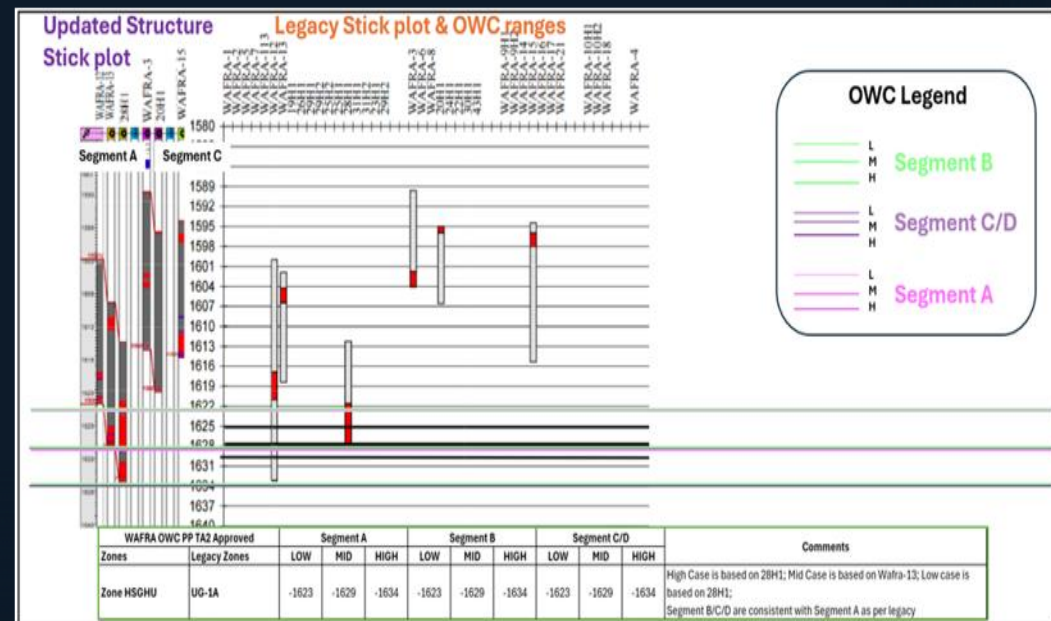
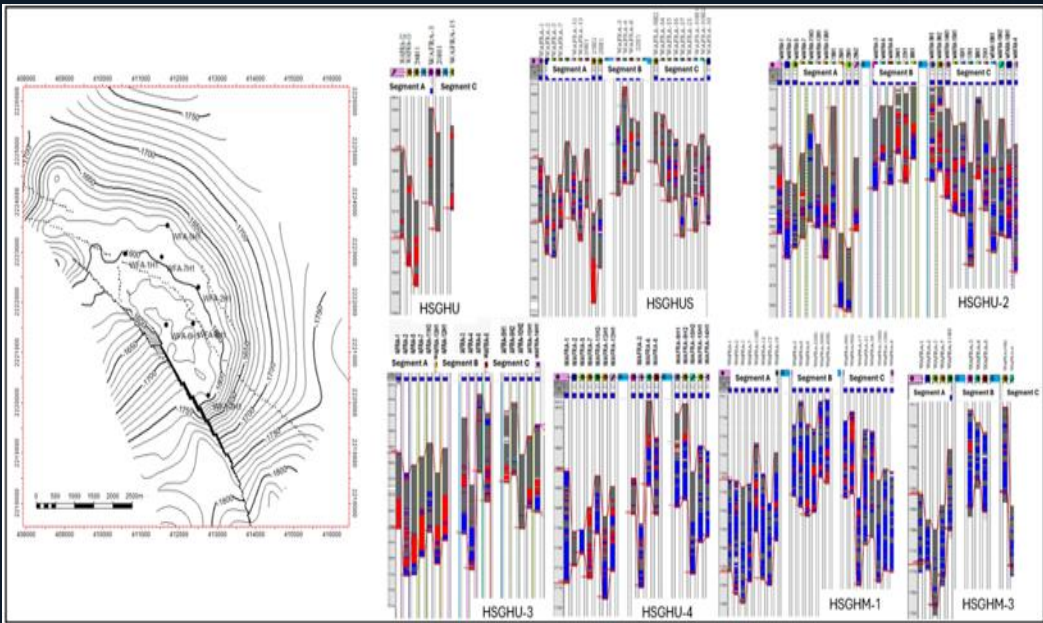
- Co-kriged with Porosity
- FMI-derived Variograms
- Depositional Control



NTG & Saturation SHF BY ZONE

NTG = 1 for Clean/Shaly Sandstone (reservoir); **NTG = 0** for Shale, Paleosol, and Carbonate. Saturation-Height Functions applied per zone from the 2010/11 DSC study.

- NTG Binary from Facies
- SHF per Zone
- 2010/11 DSC Study



FLUID CONTACT STICK-PLOT REVIEW — WAFRA FIELD

STATIC MODEL UPDATE

Fluid Contact Analysis & Saturation Modelling



STICK-PLOT METHODOLOGY

Fluid contacts revised using stick-plots of early production wells, integrating production data, pressure data, and log interpretations for a comprehensive contact determination.



7 ZONES UPDATED — CONTACT COMPILATION

HSGHU / UG-1a

HSGHUS / UG-1b

HSGHU-2 / UG-2

HSGHU-3 / UG-3

HSGHU-4 / UG-4

HSGHM-1 / MG-1a

HSGHM-3 / MG-1b



SATURATION-HEIGHT FUNCTIONS (SHF)

SHF derived from the 2010/11 DSC study applied by zone to model fluid distribution and water saturation profiles across all reservoir intervals.



VALIDATION APPROACH

Fluid distribution validated using a water saturation (S_w) cutoff of 30% applied to early production wells, confirming contact accuracy across all updated zones.

Volumetric Analysis & STOIIP Uncertainty

Three scenarios evaluated — base case [CONF.] MMm³ | Wafra Field, PDO TA2 Assessment



STOIIP values in MMm³ | Scenario 2 (PP TA2 Contacts) pending final contact update — not shown

Key Findings

BASE CASE STOIIP

Scenario 1 mid-case: **[CONF.] MMm³** with estimated RF ~[CONF.], smaller than ARPR 1P of [CONF.] MMm³ at RF ~[CONF.].

FLANK SCENARIO RANGE

Reverted Flank scenario yields **[CONF.–CONF.] MMm³**, comparable to ARPR 2P of [CONF.] MMm³ at RF ~[CONF.].

UNCERTAINTY DRIVER

Most wells drilled at crest; **limited flank well control** introduces significant volumetric uncertainty in flank areas.

ENDORSEMENT SCOPE

PDO TA2 assurers endorsed approved structure scenarios for **ML workflow only** — not for development planning or STOIIP reporting.

Note: Scenario 2 (Approved Structure + PP TA2 Contacts) incorporates updated contacts and structure; final values pending. All scenarios subject to PDO TA2 endorsement limitations.

Model Update Conclusions & Key Outcomes

Model Refinement Achieved

The updated static model delivers a refined, accurate representation of the Wafra Field subsurface. The Petrel deck was updated as a no-cost, out-of-scope activity, ensuring the model reflects the latest structural and fluid contact interpretations.

- Petrel Updated
- No Additional Cost

UG2 & UG3 as Key Flow Units

UG2 and UG3 confirmed as primary zones for BCO and infill well identification. Model enables 2 high-priority BCO candidates in UG2, plus 1 primary and 2 optional BCO candidates in UG3, alongside 8 infill opportunity assessments.

- 2 BCOs in UG2
- 3 BCOs in UG3
- 8 Infill Targets

ROCM History Match Results

UG2 and UG3 were successfully history-matched in the ROCM dynamic model. UG1 and MG zones were excluded from the match due to insufficient MOIP, limiting their role in the current development planning workflow.

- UG2 Matched ✓
- UG3 Matched ✓
- UG1 & MG Excluded

Scope & PDO TA2 Endorsement

PDO TA2 assurers endorsed the approved structure group of scenarios for ML-driven workflow execution only — not for development planning or STOIIP reporting. Flank areas flagged as a priority for future appraisal due to limited well control.

- ML Workflow Only
- Flank Appraisal Needed

 **Key Takeaway:** The model update successfully enables BCO and infill identification in UG2 & UG3. PDO TA2 endorsement is scoped to ML workflow execution; further flank appraisal is required before development planning or STOIIP reporting.